

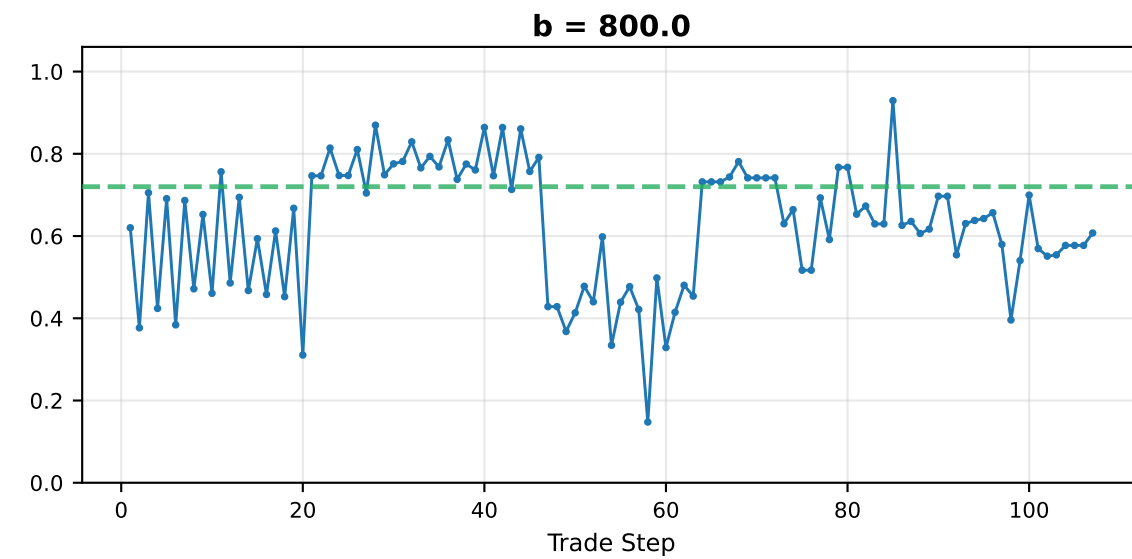
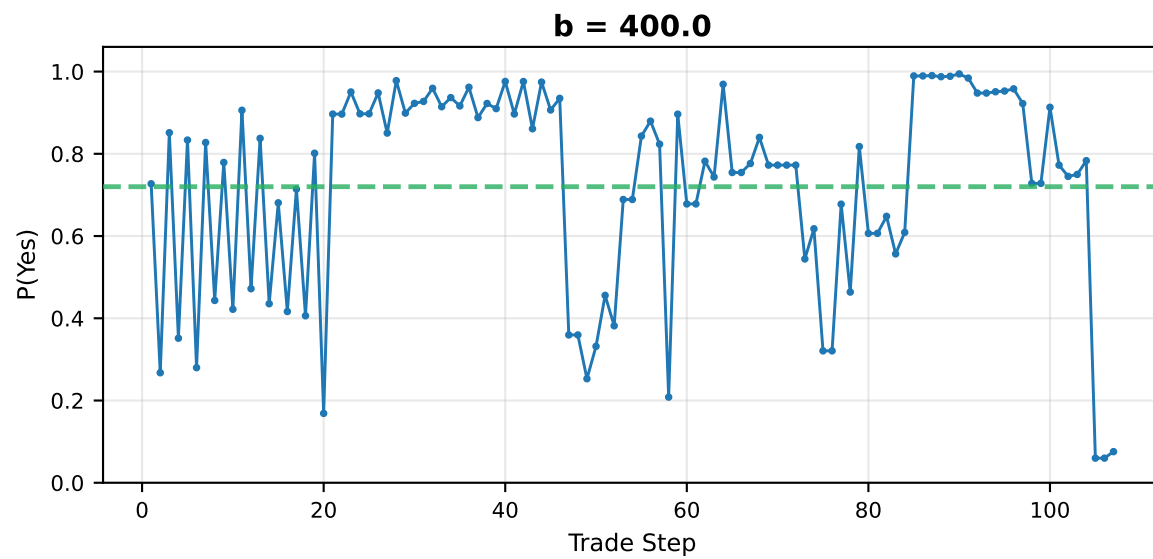
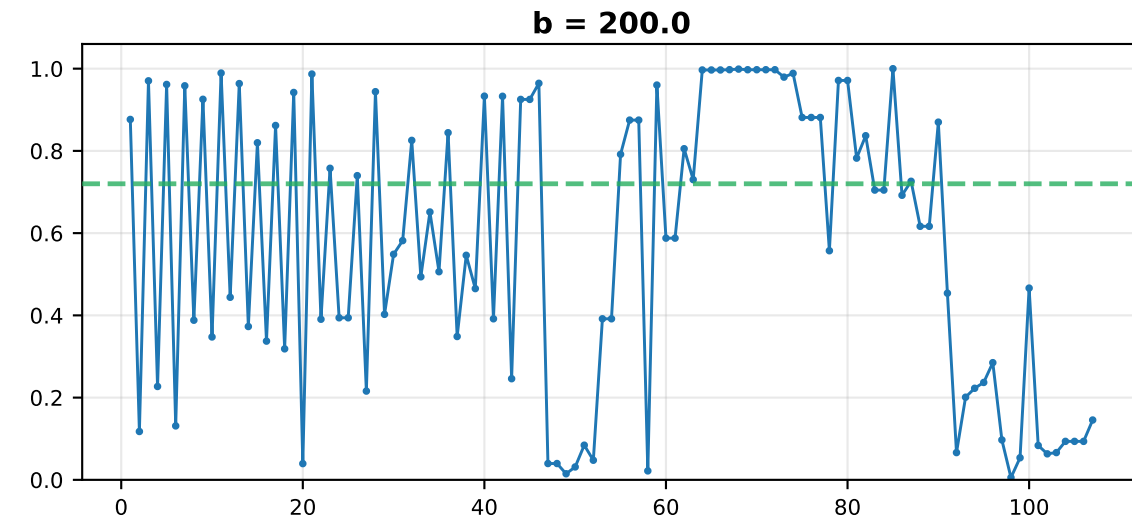
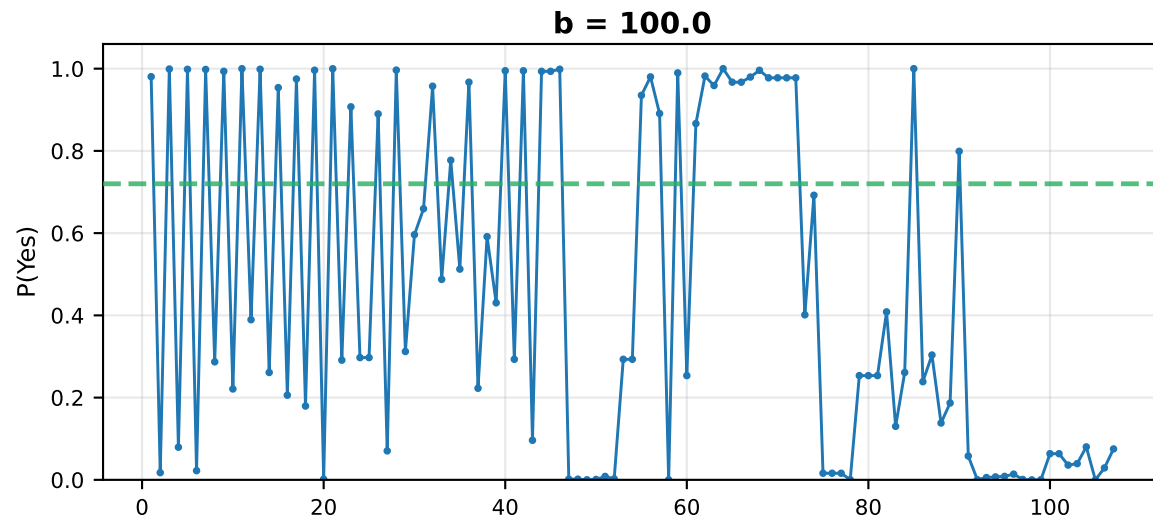
LMSR Rug Pull & Long Trend Analysis

Impact of Liquidity Parameter (b) on Price Paths

b values analyzed: [100.0, 200.0, 400.0, 800.0]

Generated from example trade histories in examples/trade_histories/

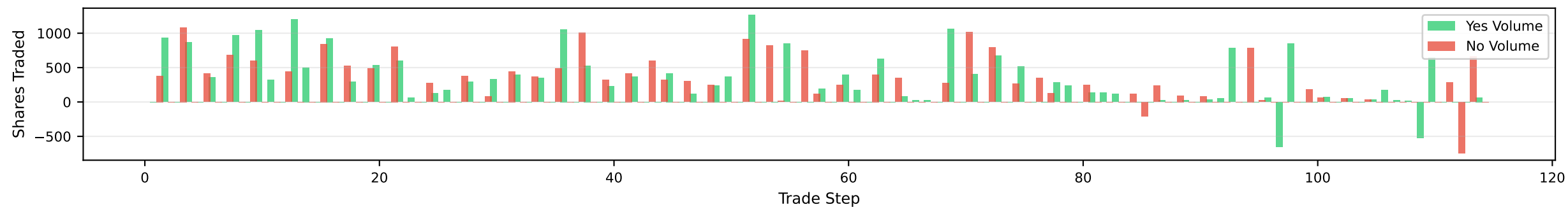
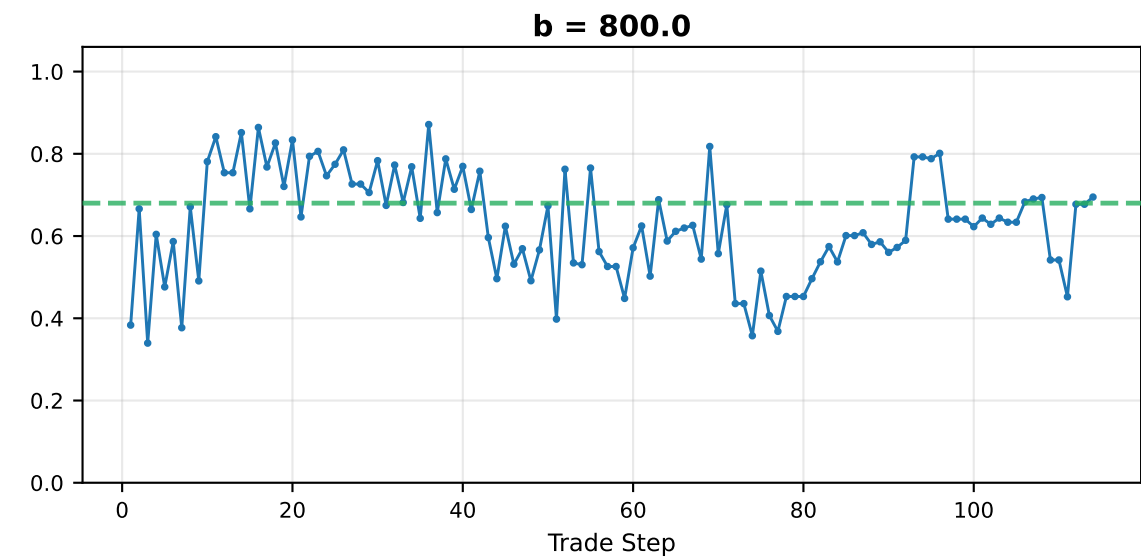
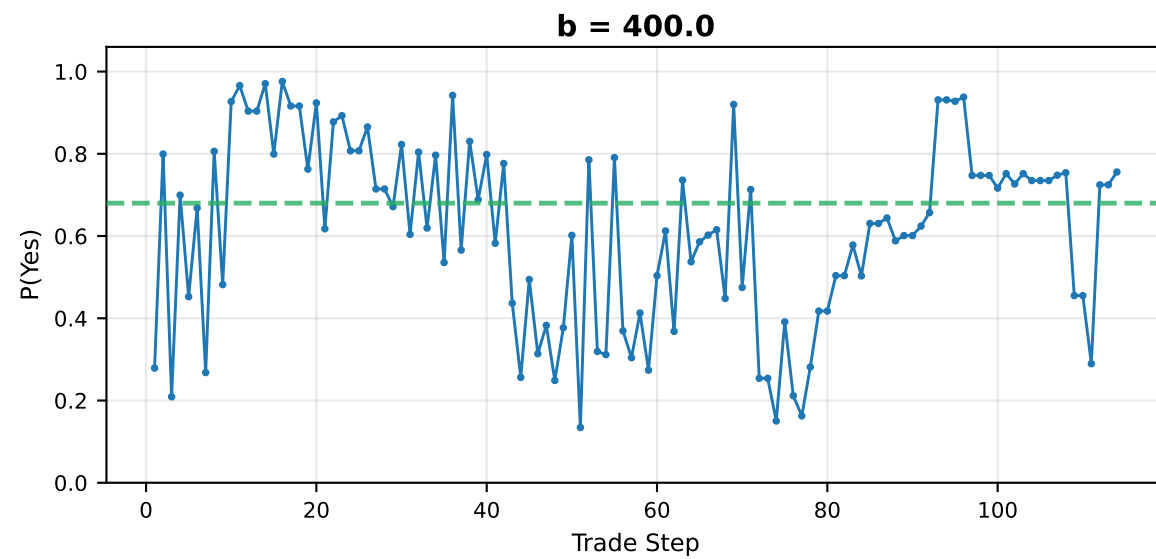
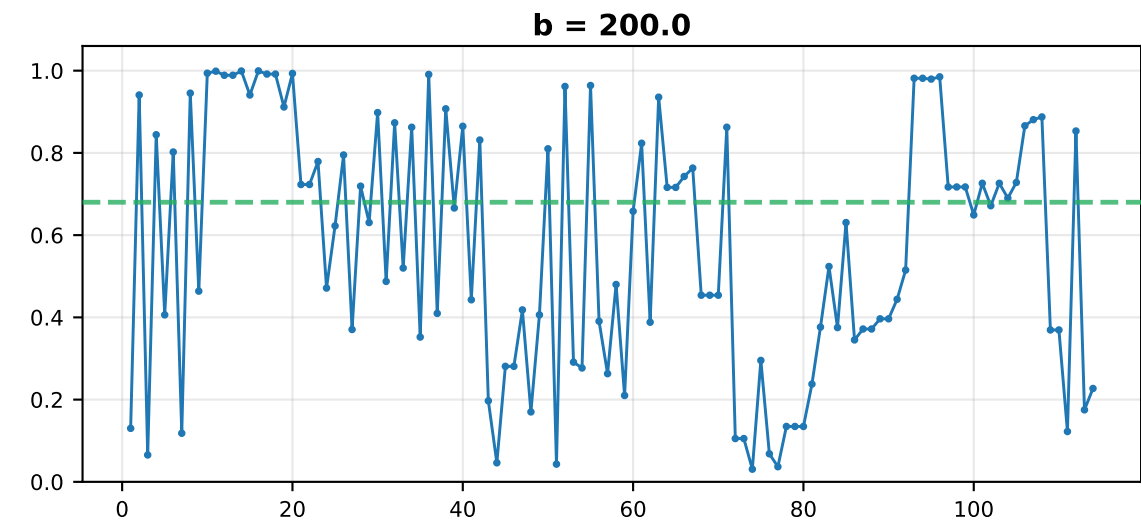
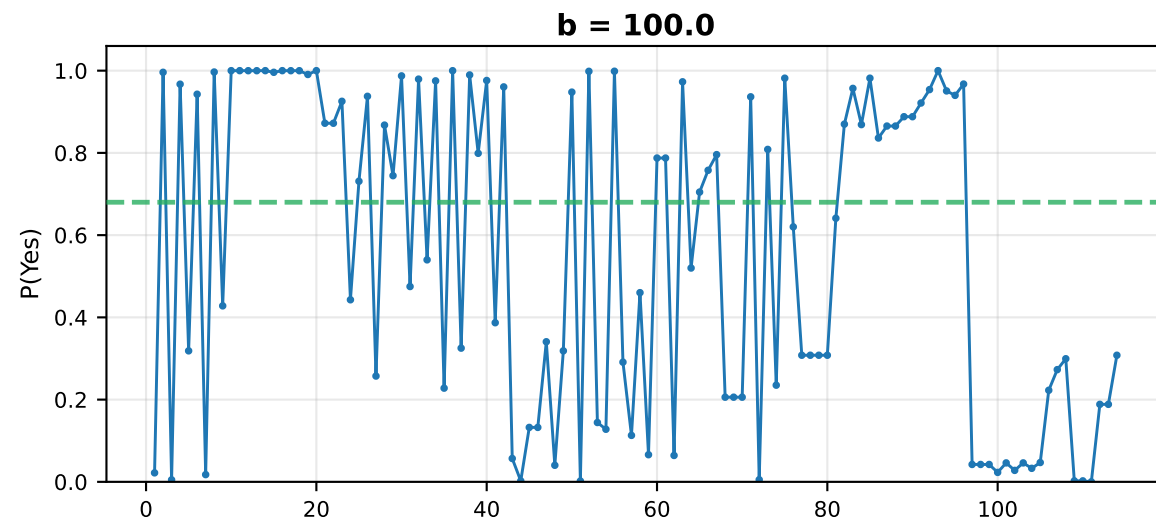
Kelly Classic Rug Pull — Price reaction per b value



Kelly Classic Rug Pull — Summary Statistics

b	QV	TV	Roughness	Step Vol	Max Step	Final p-0.5
100.0	29.2557	40.0533	7.405	0.5254	0.9981	0.4248
200.0	17.7022	30.8976	7.344	0.4087	0.9474	0.3545
400.0	7.0234	17.5802	6.634	0.2574	0.7279	0.4243
800.0	2.5216	11.8765	7.479	0.1542	0.4359	0.1074

Kelly Long Gradual Trend — Price reaction per b value

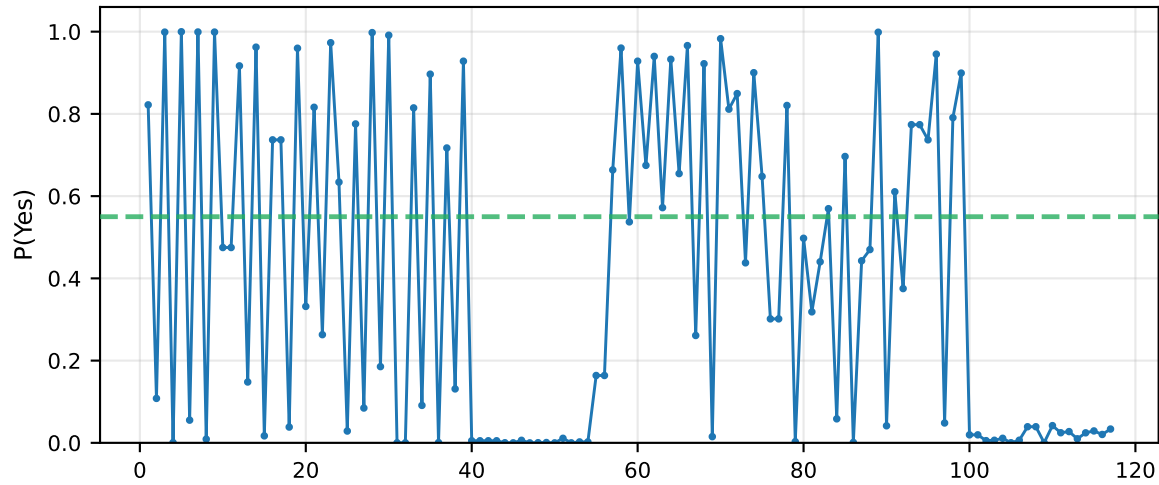


Kelly Long Gradual Trend — Summary Statistics

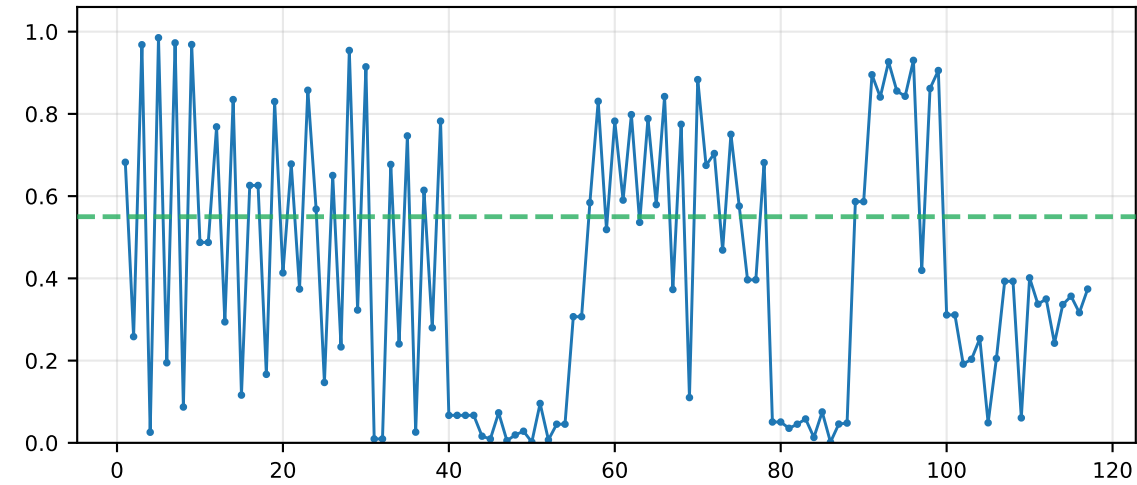
b	QV	TV	Roughness	Step Vol	Max Step	Final p-0.5
100.0	23.6134	35.4500	7.295	0.4571	0.9964	0.1921
200.0	14.1278	28.2150	7.507	0.3536	0.9185	0.2729
400.0	6.0221	18.6769	7.611	0.2309	0.6510	0.2558
800.0	1.9012	10.7412	7.790	0.1297	0.3645	0.1948

Kelly High-Activity Noisy Market — Price reaction per b value

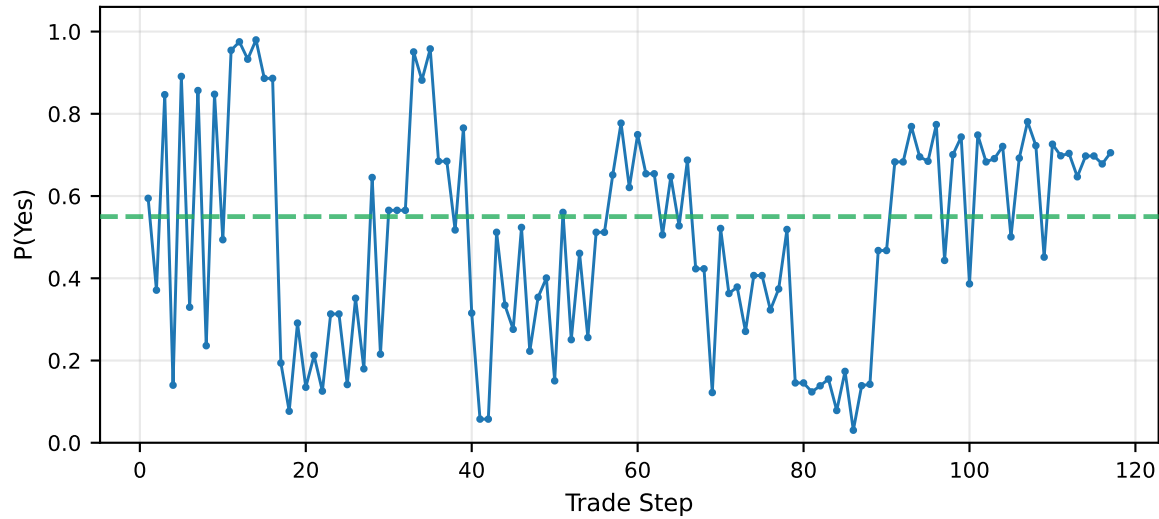
b = 100.0



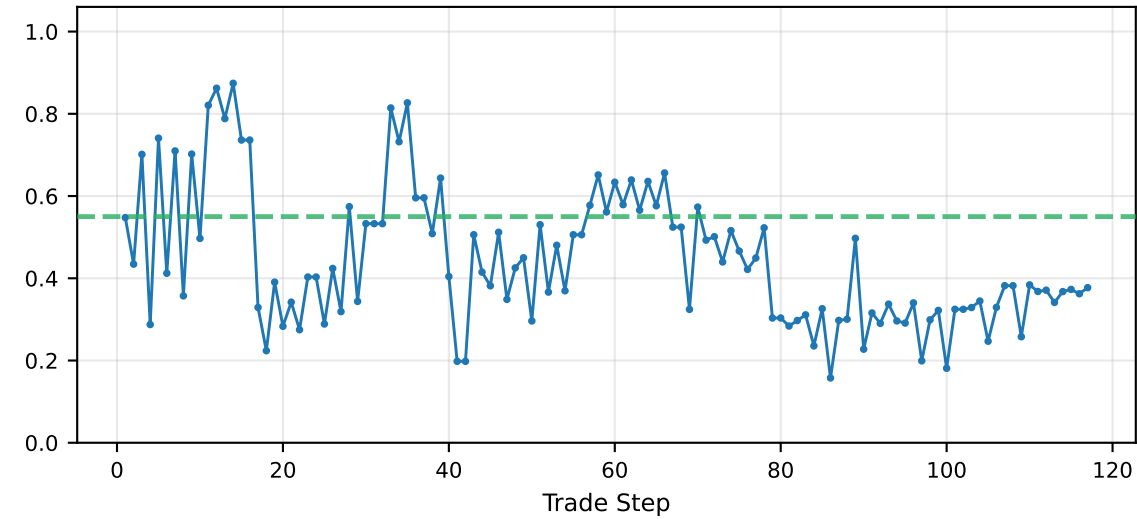
b = 200.0



b = 400.0



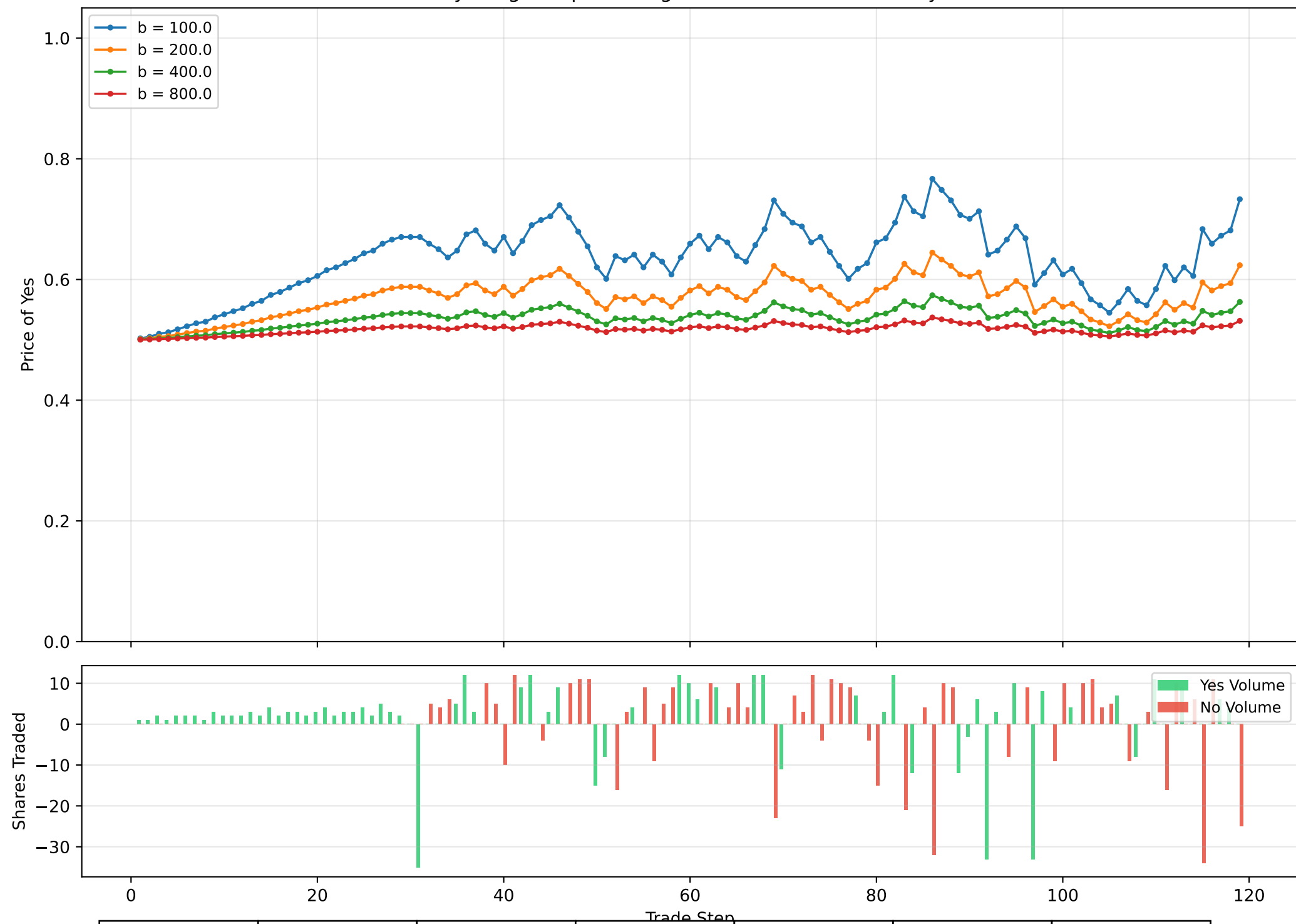
b = 800.0



Kelly High-Activity Noisy Market — Summary Statistics

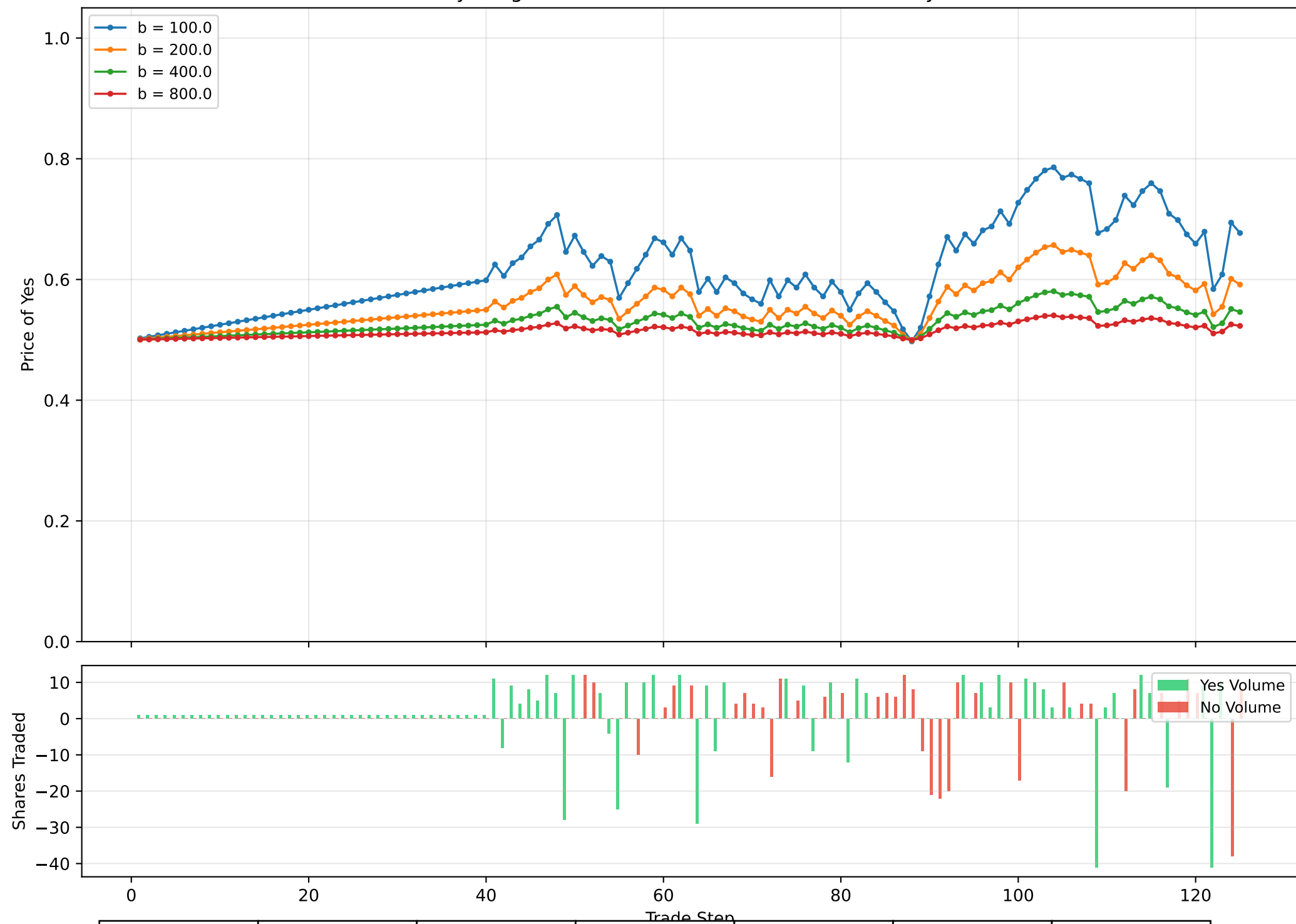
b	QV	TV	Roughness	Step Vol	Max Step	Final p-0.5
100.0	33.6572	46.7846	8.064	0.5387	0.9991	0.4664
200.0	17.9963	33.2698	7.843	0.3939	0.9594	0.1260
400.0	7.3902	21.0828	7.755	0.2524	0.7509	0.2053
800.0	2.6527	12.9359	7.942	0.1512	0.4533	0.1228

Very Long Pump and Rug — Price Path + Volume by b



b	QV	TV	Roughness	Step Vol	Max Step	Final p-0.5
100.0	0.0614	2.0933	8.451	0.0228	0.0776	0.2330
200.0	0.0184	1.1346	8.374	0.0125	0.0416	0.1236
400.0	0.0048	0.5798	8.359	0.0064	0.0212	0.0628
800.0	0.0012	0.2914	8.348	0.0032	0.0106	0.0315

Very Long Gradual Trend — Price Path + Volume by b



b	QV	TV	Roughness	Step Vol	Max Step	Final p-0.5
100.0	0.0764	2.1749	7.870	0.0248	0.0950	0.1770
200.0	0.0223	1.1717	7.851	0.0134	0.0503	0.0915
400.0	0.0058	0.5977	7.842	0.0068	0.0255	0.0461
800.0	0.0015	0.3010	7.839	0.0034	0.0128	0.0231

Key Observations

- Lower b values cause much larger and faster price swings.
- In rug-pull scenarios, low b makes the price run much higher before the dump.
- High b markets are far more resilient to large single trades (less slippage).
- In long gradual trends, low b can push prices close to 1.0, while high b stays moderate.
- Rug pulls with fake volume (coordinated accounts) are more effective at low b .

Recommendation for internal tools (with Kelly-sized bets):

- $b = 100\text{--}200$: noticeable impact, still educational volatility
- $b = 400+$: much more stable, good for serious forecasting markets
- $b \geq 500\text{--}800$: very deep markets (e.g. Experts vs Punters style)
- Only use $b < 50$ when you specifically want extreme sensitivity.